Phenoneurogenomic Decomposition of Diagnosis and Sex in Autism Spectrum Disorder

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Introduction

Autism Spectrum Disorder (ASD) is now appreciated as likely representing a collection of developmental conditions whose outward behavioral manifestation belies defies a single neurological of genetic source. Children with ASD present a diagnostic ratio of 5:1 in boys (♂) vs. girls (♀), have been reported to display altered connectivity[1], and the syndrome has been associated with <400 genes. Our study, as part of the National Institutes of Health (NIH) Autism Center of Excellence (ACE) Program, sought to explore each of these domains and their interactions via comprehensive phenomic assessment, structural and connectomic human neuroimaging, and whole genome sequencing. The principal goal of this investigation was to identify the sources of sex-specificity in ASD vs. Typically Developing (TD) children.

Methods

Neuropsychological, neuroimaging, and genetics data were acquired from four research centers (Yale, UCLA, Harvard, and U. Washington). Cohort membership was based upon the results of the revised Autism Diagnostic Interview (ADI-R) and the Autism Diagnostic Observation Schedule (ADOS). N=111 subjects with ASD (49 ♂, 62 ♀) and N=115 typically developing (TD) subjects (57 ♂, 58 ♀), between the ages of 8 and 12, were included for domain-specific analysis by diagnosis-by-sex.

Neuropsychological assessments: A battery of formal neuropsychological measures was administered to each subject. These included the sub-scales of: the DAS; Vineland; SRS; CELF; RBSR; SCQ; BRIEF; and CBCL for a total of 47 predictor variables across the four groups. Multivariate linear discriminant analysis (LDA) was performed indicating two principle axes of discrimination accounting for 85% and 12% of the variance, respectively ($p<0.001$ by Wilk's Lambda).

Neuroimage acquisition and processing: MRI images (MPRAGE, BOLD-EPI, and DWI) were acquired on a Siemens Magnetom TrioTim or Siemens Magnetom PRISMA scanner with magnetic field strength of 3T via a standardized acquisition protocol across sites. Image data preprocessing was performed using the FSL and parcellation of the brain was achieved using FreeSurfer 5.3 against the atlas of Destrieux *et al*. Connectograms were computed as described previously[2]. All results were normalized by intra-cranial volume. Independent *t*-tests were systematically performed to assess differences in group means for each of those regional values with resulting p-values corrected for multiple comparisons using the Benjamini-Hochberg algorithm.

Genome "Hotspot" Analysis: Genome sequencing (WGS; Illumina Infinium Omni2.5-8 Kit) was performed on blood samples obtained from the ASD ♂'s and ♀'s, their parents, and unaffected siblings. VCF files for each subject were compared against the 1,000 Genomes[3] reference genome. Chromosomes were each divided into 10,000 segments and used to compute mutation densities. Mutation frequencies were then averaged by group. Karyotype maps (**Figure 3**) illustrate mutation frequencies in the 75%'ile. Statistical differences were assessed between groups to determine mutation loci present in ASD ♂'s not present in ASD ♀'s, and *vice versa*. Loci were systematically compared to the locations of known genes (UCSD Genome Browser), their exomic sections, and prior association with ASD, etc.

Results

Neuropsychological assessments: LD1 was associated with the clear separation of ASD vs. TD subjects (**Figure 1**). LD2 was associated with the interaction between ASD ♂ and ♀ vs. TD ♂ and ♀, where the ASD groups showed clear separation whereas the TD groups did not. The BRIEF sub-scales[4] were particularly relevant on LD2. Ten-fold cross-validation showed a *posteriori* classification accuracy = 64.95%

Neuroimaging: Analyses found no significant differences in the regional morphometry of gray matter structures between sexes or diagnostic group (Figure 2) and no differences in FA or CD connections between ASD and TD cohorts. Future work will investigate the differences in local and global connectivity metrics between sexes and diagnostic groups.

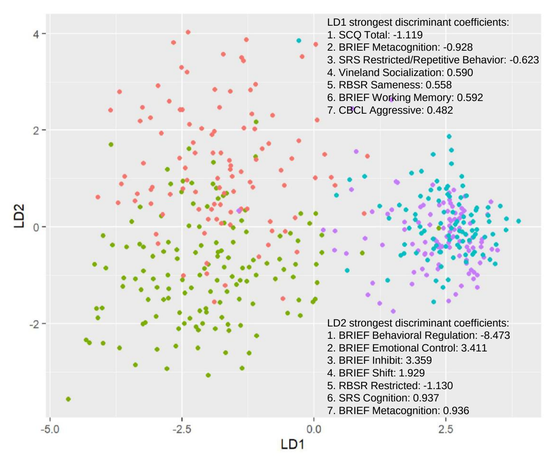

Figure 1. LDA between the ASD and TD male and female cohorts. LD1 showed separation based on diagnosis, while LD2 represented the interaction of diagnosis by sex. Red=ASDF; Greenn=ASDM; Blue=TDF; Purple=TDM.

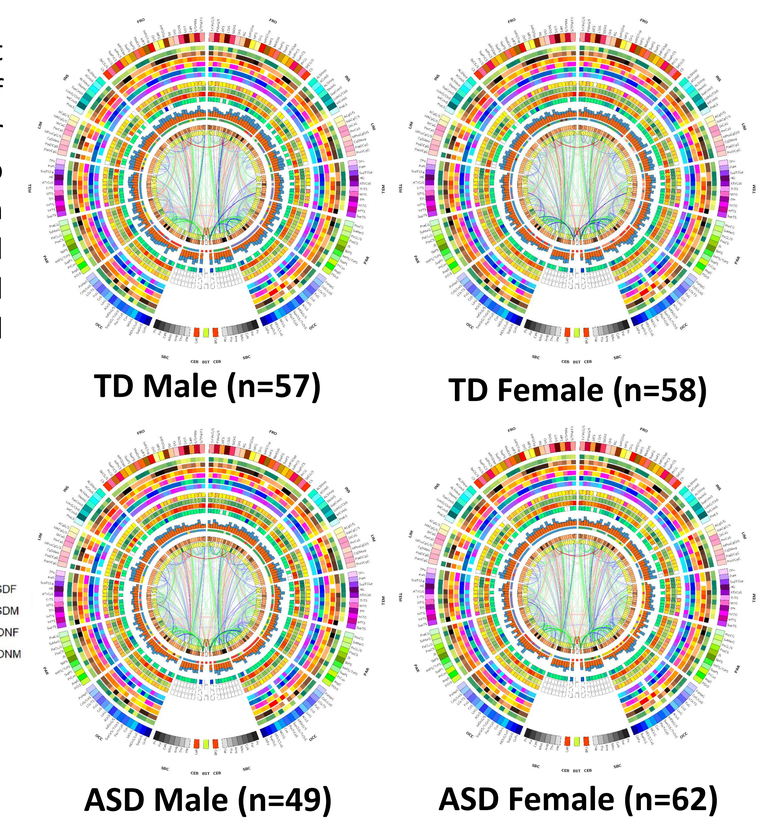

Figure 2. Average connectograms associated with each cohort, reflecting (outside-in) various brain atlas reference spaces, mean cortical morphometrics, functional connectivity metrics, degree of white matter connectivity by region, and interregional connectivity coded by density (line opacity) and average fascicular fractional anisotropy (color; red=high, green=mid, blue=low).

Genetics: Our mutation density analysis identified the following genes in chromosome-1, which were already implicated in ASD: NEGR1, IQGAP3, DISC1 and RGS7 (figure 3A). The following loci were found to be sex-specific "hotspots": Segment [93.70, 93.72]M in chr-9, segments [2.55, 2.57]M, [24.05, 24.07]M, [55.18, 55.20]M, [118.46, 118.48]M, and [155.94, 155.96]M in chr-X, and segments [11.63, 11.65]M and [20.25, 20.27]M in Chr-Y (**Figure 3B**). No sex differences were noted on Chromosome 17[5].

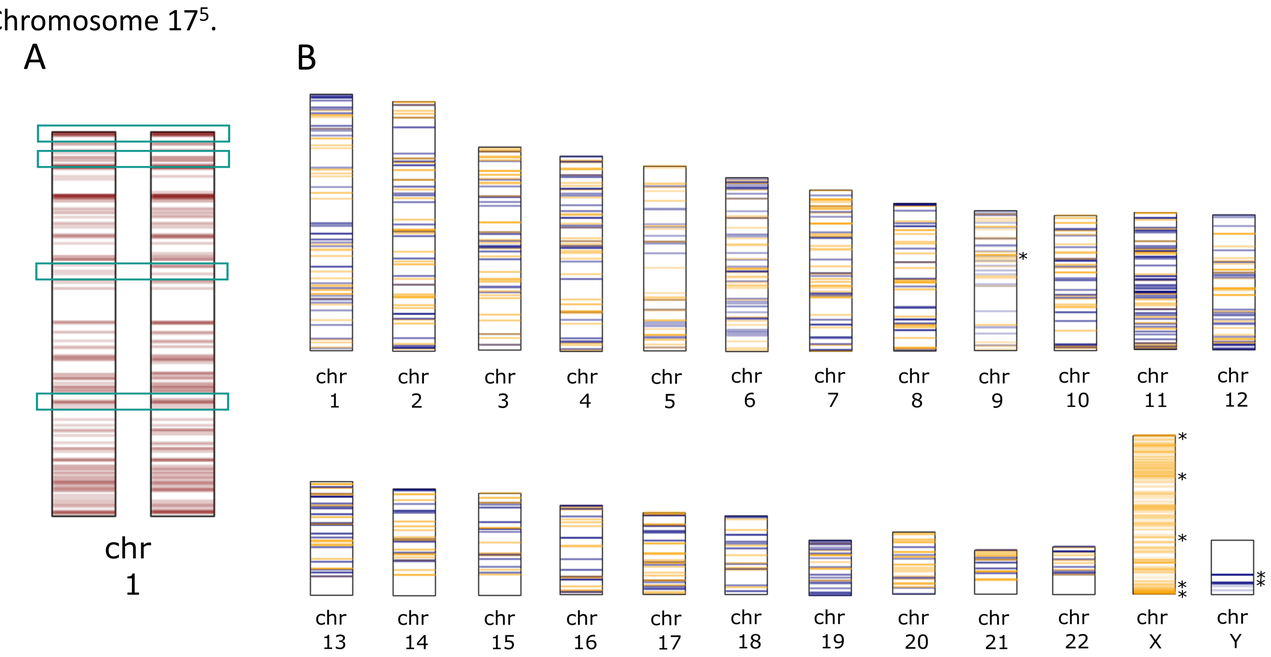

Figure 3. A: An example of hot-spots of mutation densities identified in male (left) and female (right) chromosome-1. Highlighted loci indicate the approximate exonic regions of genes previously associated with ASD. **B.** Sex-specific mutation frequency "hotspot" karyotype mapping between ASD ♂ and ♀. T-statics of differences between males and females are mapped for each chromosome segment. Orange and blue denote higher mutation densities in females and males, respectively. * denotes statistically significant differences

Discussion

Findings of a multivariate phenomic diagnosis-by-sex interaction, driven largely by assessment subscales of executive function and self-regulation, suggest that these cognitive domains may be particularly sensitive to sex-differences in ASD unlike as in TD children[3]. No volumetric differences in cortical morpometry suggest that previous findings may be influenced by intra-cranial volume variation between ♂ and ♀, respectively. Differences in network theoretical metrics of functional connectivity necessitate further exploration. Gene loci previously linked to ASD, show variable mutation rates between ♂ and ♀, suggestive of differential genetic influences. Our study confirms the contention that ASD is sexual dimorphic which may hinder effective and consistent definition and diagnosis.

References

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